

01 Overall Spatial Structure

One axis · Three zones · Two wings · Six nodes · Five layers

This proposal reads the AI innovation belt not as a showcase axis of landmark buildings, but as a network of urban public infrastructure — connectable, exchangeable, incrementally buildable, sustainably operable, and replicable — organising the railway heritage park, universities, technology parks, communities, and ecological spaces.

Three Litmus Questions (no answer, no build)

- 1 Connection: which specific rupture does it stitch — the two sides of the railway, inside and outside institutional walls, old and new communities?
- 2 Exchange: what resources does it set genuinely flowing between which actors — space, equipment, materials, knowledge, data?
- 3 Growth: what interfaces does it leave for the next stage of expansion, adjustment, or exit?

Jingzhang Growth Section (basic spatial unit)

- Above ground Adaptable R&D units — demountable envelopes, independent MEP, reconfigurable in 3–5 years
Shared courtyards — inter-institutional exchange: equipment, samples, courses, joint experiments
Open ground floors — publicly accessible: exhibition, courses, community services, small retail
Ecological restoration plots — productive landscape: habitat monitoring, stormwater, material trials
Railway heritage park — continuous public base: walking/cycling main line, heritage narrative, observation
Background Urban observation and operations network — sensing, data archiving, maintenance routes

Every interface in the section is an 'exchange'; segments take different slices — Zhongzhiyuan shows all five layers, AI Origin strengthens courtyard–ground floor, Dazhongsi strengthens ground floor–park.

01 Site Overview: Issues & Opportunities

Jingzhang Adaptive Commons · Concept V0.2 | Coordinated Research Area 43.6 km² · Overall Design Area 11.4 km² (both provisional outlines)

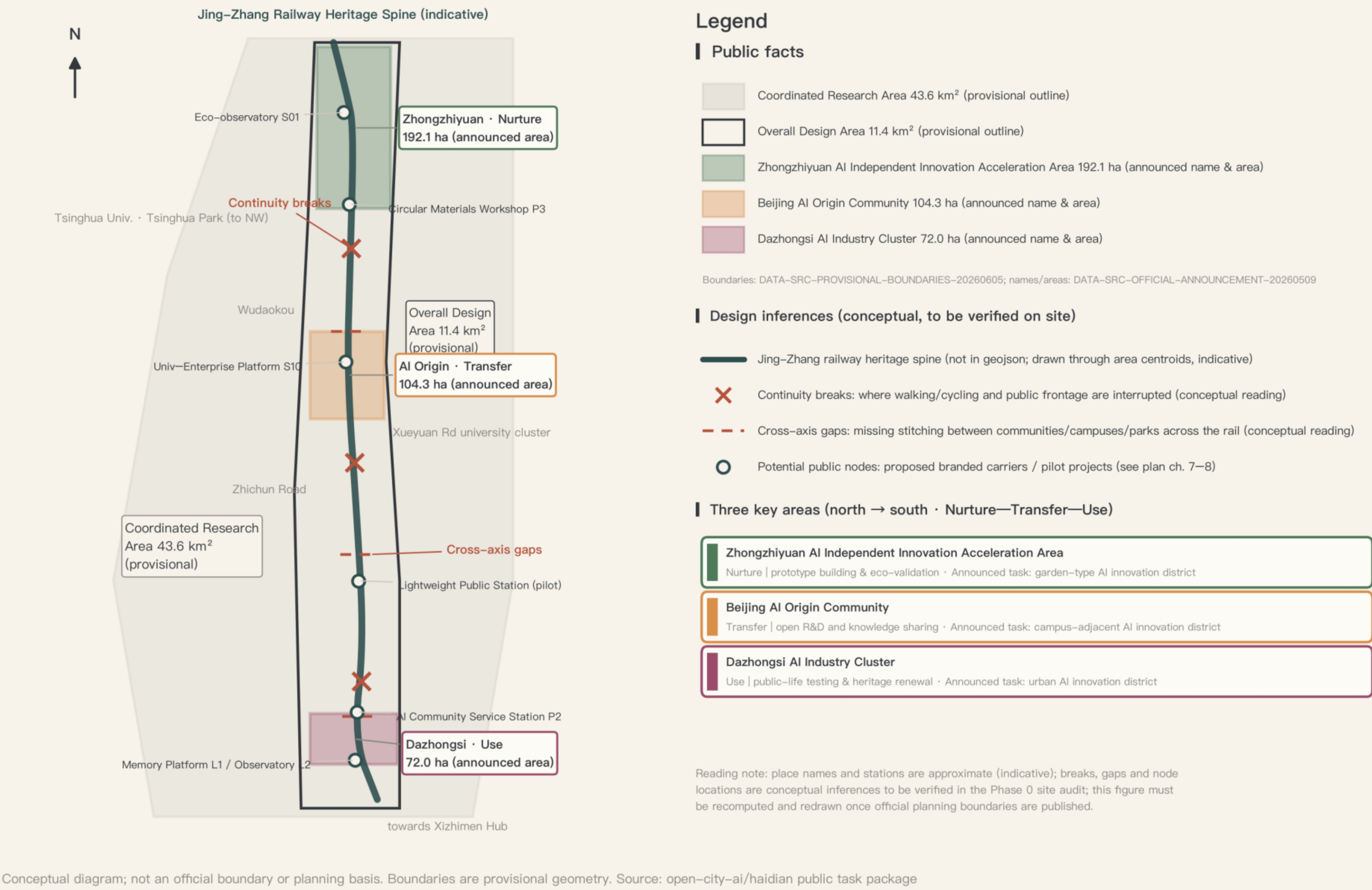


Fig. 1 Site overview, issues & opportunities (boundaries are provisional constraints, not official red lines)

02 Overall Structure & Functional Organization

One public spine · three key areas (Nurture—Transfer—Use—Return) · six distributed node types · five layered networks | structural diagram, not geographically precise

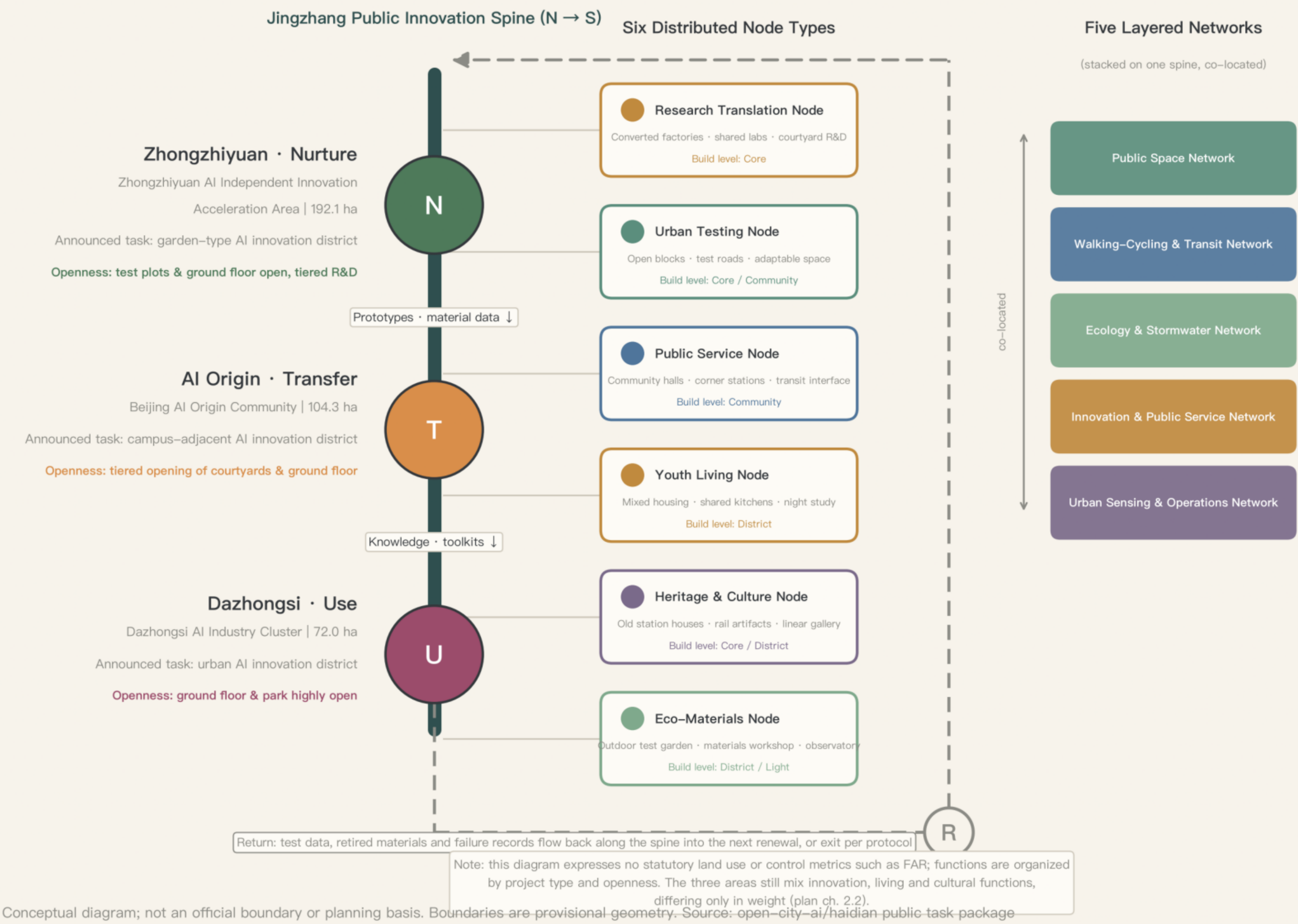


Fig. 2 Overall structure and functional organisation: one axis, three zones (Nurture–Transfer–Use–Return), two wings, six node types, five-layer network

Area (announced)	Role	Core proposition
Zhongzhiyuan ~192.1 ha	Nurture: prototyping & validation	Test plots, material trials, and prototyping as the everyday ground floor
AI Origin Community ~104.3 ha	Transfer: open R&D	University knowledge production mutually visible and usable with community life
Dazhongsi Cluster ~72.0 ha	Use: public-life testing	AI and spatial prototypes tested in the most everyday, urbanised environment

Test data, retired materials, and failure records flow back (Return) into the next round of renewal or exit by agreement.

Six node types & five-layer network

Six node types: research-transfer (core) / urban-experiment (core, community) / public-service (community) / youth-life (district) / heritage-culture (core, district) / ecology-materials (district, lightweight).
Five-layer network: public space, walking-cycling-transfer, ecology-stormwater, innovation and public service, urban observation and operations — superimposed on the axis; one lateral stitching point solves crossing, stormwater retention, services, lighting, and interpretation at once rather than five separate sectoral projects.
Two wings: the Zhongguancun Technology Services Wing (west, an interface belt for technology transfer and producer services) and the Xiaoye River Scenario Enablement Wing (east, a blue-green corridor and scenario-testing belt), each joining the axis at city-level stitching points; wing boundaries are provisional expressions.

Conceptual land–use model (from provisional geometry; not statutory)

Category	Area (ha)	Share
Green & water	319.0	27.96%
Education & research	243.3	21.3%
Industry & R&D	206.7	18.1%
Residential support	148.6	13.0%
Public space	125.4	11.0%
Commercial services	98.2	8.6%

Shares are estimated against the recalculated area of 1,141.28 ha (roads and other unclassified parts excluded); all values are conceptual design–model figures (low confidence) and must be recalculated once regulatory planning is published. FAR, height, coverage, and setbacks remain to be confirmed — no numerical conclusions.

Retain–renovate–demolish strategy

Identify retainable buildings, mature trees, existing public spaces, and social networks before deciding on new construction. Factory buildings, warehouses, and large spans are prioritised for adaptive reuse as innovation halls and workshops; low–quality structures blocking public connectivity enter demolition evaluation; the rest are mainly renovated. The 14 building footprints are conceptual; no per–building conclusions before parcel, ownership, and survey data arrive.

Seven renewal–strategy principles

Existing–stock first · continuous base · distributed configuration · functional reciprocity · phased implementation · circular construction · feedback correction. All intensity controls (FAR, height, coverage, setbacks) remain to be confirmed until regulatory planning materials arrive; land–use ratios and intensity indicators draw no statutory conclusions before data is in place.

Four building prototypes

A · Retrofitted Innovation Hall

Adaptive reuse of factories/large spans; a central shared hall links lab, roadshow, exhibition, and making; independent MEP and movable partitions adapt to team sizes.

B · Courtyard R&D Unit

Small R&D buildings around shared courtyards; shared meeting, equipment, and sample spaces below, divisible units above; corridors link campus, park, and axis.

C · Permeable Public Block

Ground floors with multiple entrances, shallow depths, high visibility; services, food, learning, sports, and culture along the axis; internal paths continuous with community streets.

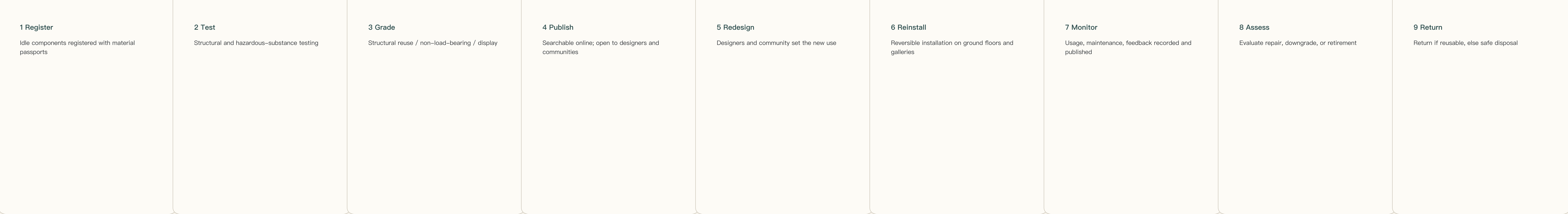
D · Lightweight Public Station

Demountable structure, standardised interfaces, minimal foundations; services, observation, micro–exhibitions, rest, repair; piloted first, adjusted by usage data.

Character control

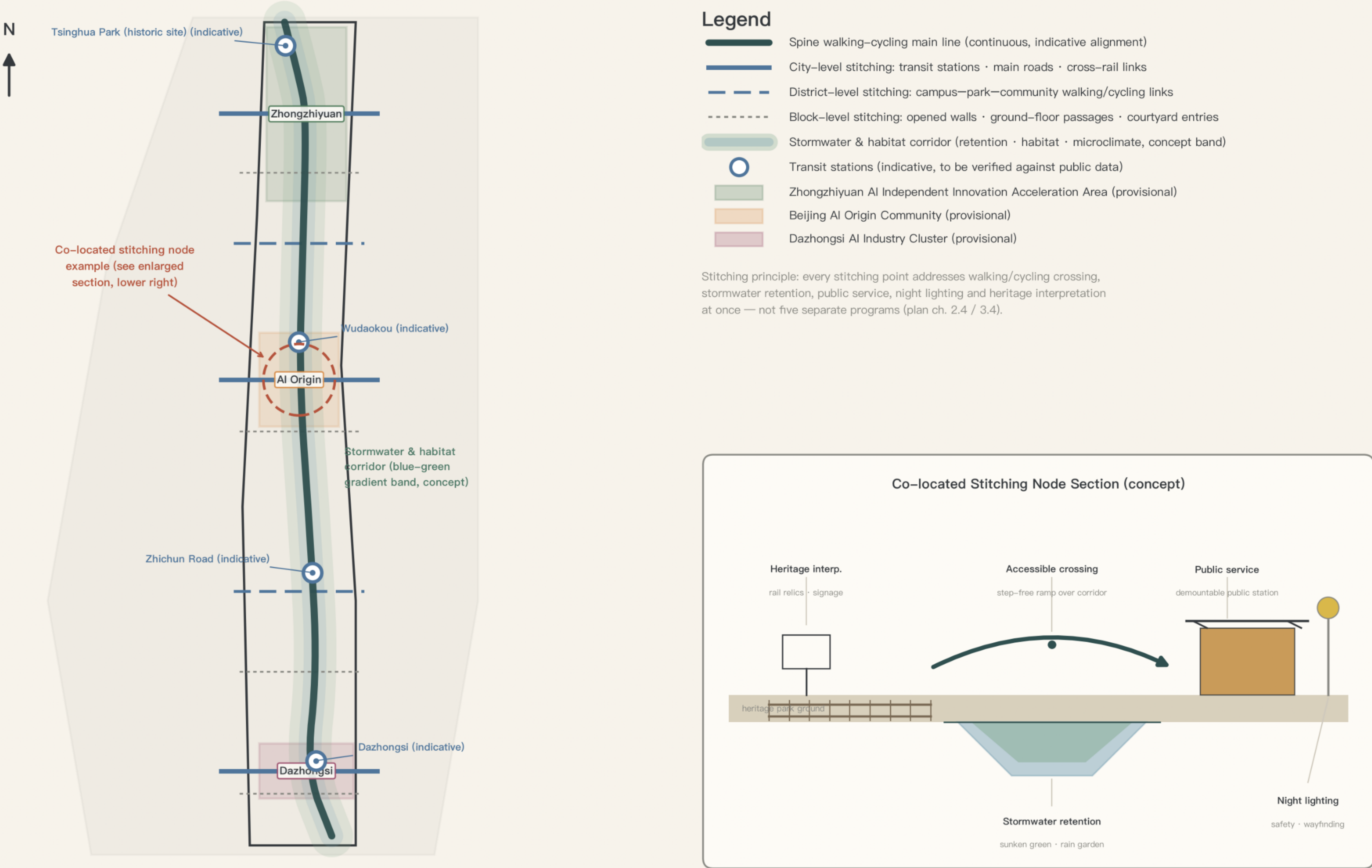
The proposal rejects over–curved 'natural–form' expression and adopts an architectural language suited to the Jing–Zhang Railway and Zhongguancun innovation culture — clear structural grids and extensible units; linear platforms, canopies, pergolas, and cross–line structures; brick, steel, timber, and reused components forming a legible old–and–new material relationship; courtyards, lanes, grey spaces, and public ground floors forming a porous interface; equipment, services, and envelope layers that are maintainable and replaceable. Iconicity comes from public activity and structural order, not from a one–off sculptural skin.

The nine–step cycle of one component (E04 platform × P3 workshop)



04 Walking–Cycling, Transit & Blue–Green Networks

Spine walking–cycling main line · three levels of cross–stitching · transit stations (indicative) · stormwater & habitat corridor | based on provisional geometry



Conceptual diagram; not an official boundary or planning basis. Boundaries are provisional geometry. Source: open–city–ai/haidian public task package

Fig. 4 Walking, cycling, transfer, and blue–green networks (all alignments conceptual; no fabricated engineering alignments)

Judgment: the core of the transport strategy is not new corridor capacity but three levels of lateral stitching — reconnecting, tier by tier, the everyday links cut by the railway, arterial roads, and institutional walls. Road red lines, sections, traffic, parking, and utility capacity data are unavailable.

Three levels of lateral stitching

City–level: rail stations, major roads, and cross–line passages for transfers and cross–district links; both wings join the axis here.
District–level: walking/cycling passages linking campuses, parks, and communities, e.g. the innovation walking loop at AI Origin.
Block–level: opening walls, ground–floor passages, and courtyard entrances.
All stitching points are checked for accessibility continuity, shelter, nighttime safety, wayfinding, and operating responsibility.

Rail and stations

Dazhongsi: continuous four–quadrant walking connections around the station (concourse links, improved crossings, integrated platforms).
AI Origin: walking breaks between the station, campuses, and communities sorted out.
Zhongzhiyuan: station–park–axis three–line connection and last–mile walking quality strengthened.
Station–access sections and passage alignments await road red–line and metro data.

Static traffic

Parking–occupied negative spaces around Dazhongsi station gradually become public ground floors and slow–traffic space; total control, shared parking, and underground coordination await transport–sector data — no figures preset.

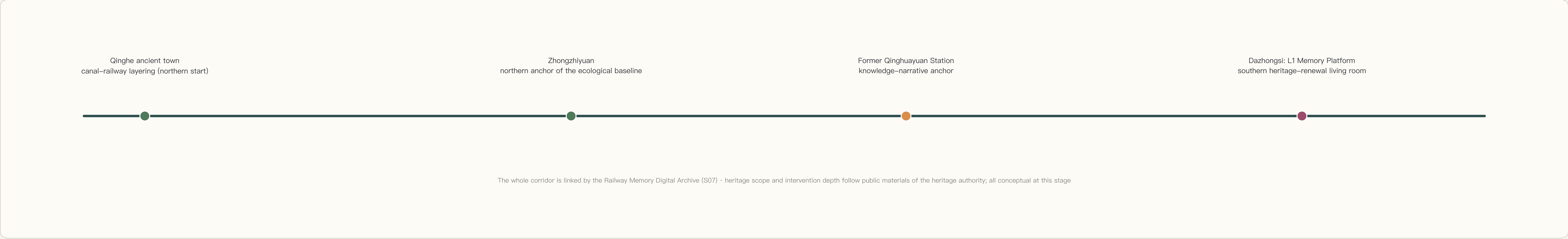
Municipal & new infrastructure

The urban observation and operations network runs corridor–wide as a background layer (sensing, data archiving, maintenance routes), sharing facilities and data specifications with scenarios such as S01 and S05; traditional utility coordination, fire protection, flood drainage, and capacity data are not public — related indicators remain to be confirmed.

Public service facilities

Public–service nodes (community halls, street–corner stations, transfer interfaces) carry education, health, government affairs, accessibility, and digital services at community level, incorporated into S03/S05 operations; existing baselines are unavailable, layouts are conceptual, and service–radius rechecks are required at the sectoral–planning stage.

Heritage narrative band (north → south)



Blue–green network

The main axis superimposes habitat, stormwater, and microclimate corridor functions; Zhongzhiyuan's restoration plots and material trial fields form the northern anchor of the corridor–wide ecological baseline; the east wing organises a waterfront blue–green corridor along the Xiaoyue River, with the avoidance extent of the Yuan city–wall relics to be confirmed with heritage–authority materials.

Under the conceptual model, green space and water cover ~319.0 ha, ~27.96% of the recalculated area within the provisional boundary — a design–model value expressing the magnitude of the blue–green skeleton, not a statutory greening rate, which remains to be confirmed.

Three types of public–space interfaces

Continuous interfaces: the walking/cycling main line, heritage display, tree shade, and nighttime lighting remain basically continuous.

Exchange interfaces: shared halls, squares, and public ground floors where campuses, parks, and communities meet the axis.

Quiet interfaces: habitat buffers, rain gardens, and low–disturbance rest areas — no activity or experiment may intrude.

Under the conceptual model, public space covers ~126.8 ha, ~11.11% of the recalculated area; publicness is guaranteed by open hours, the share of free events, and operating rules — not by area alone. Public space stays continuously open and free of charge; corridor–wide commercialisation and over–programming are avoided.

Urban character, stormwater & habitat

Architectural language: clear structural grids and extensible units; linear platform, canopy, pergola, and cross–line vocabulary; brick, steel, timber, and reused components in a legible old–new relationship; courtyards, lanes, grey spaces, and public ground floors as a porous interface; maintainable and replaceable equipment, services, and envelopes; iconicity from public activity and structural order, not a one–off sculptural skin.

Stormwater and habitat: retention and habitat creation are co–located with walking/cycling and services at the network layer, not separate sectoral projects; ecological–improvement claims rest solely on the P1 protocol's controlled monitoring — no claim of improvement without a baseline.

Still missing: surveys of existing trees, water systems, and public spaces; soil, water, habitat, and microclimate baselines; heritage protection boundaries and construction control zones.

~319.0 ha · 27.96%

Green space & water area and share (conceptual design–model value, not the statutory greening rate)

~126.8 ha · 11.11%

Public–space area and share (conceptual model value, incl. three stitching–node squares and four 12 m public ground–floor interface bands)

Statutory greening rate: to be confirmed

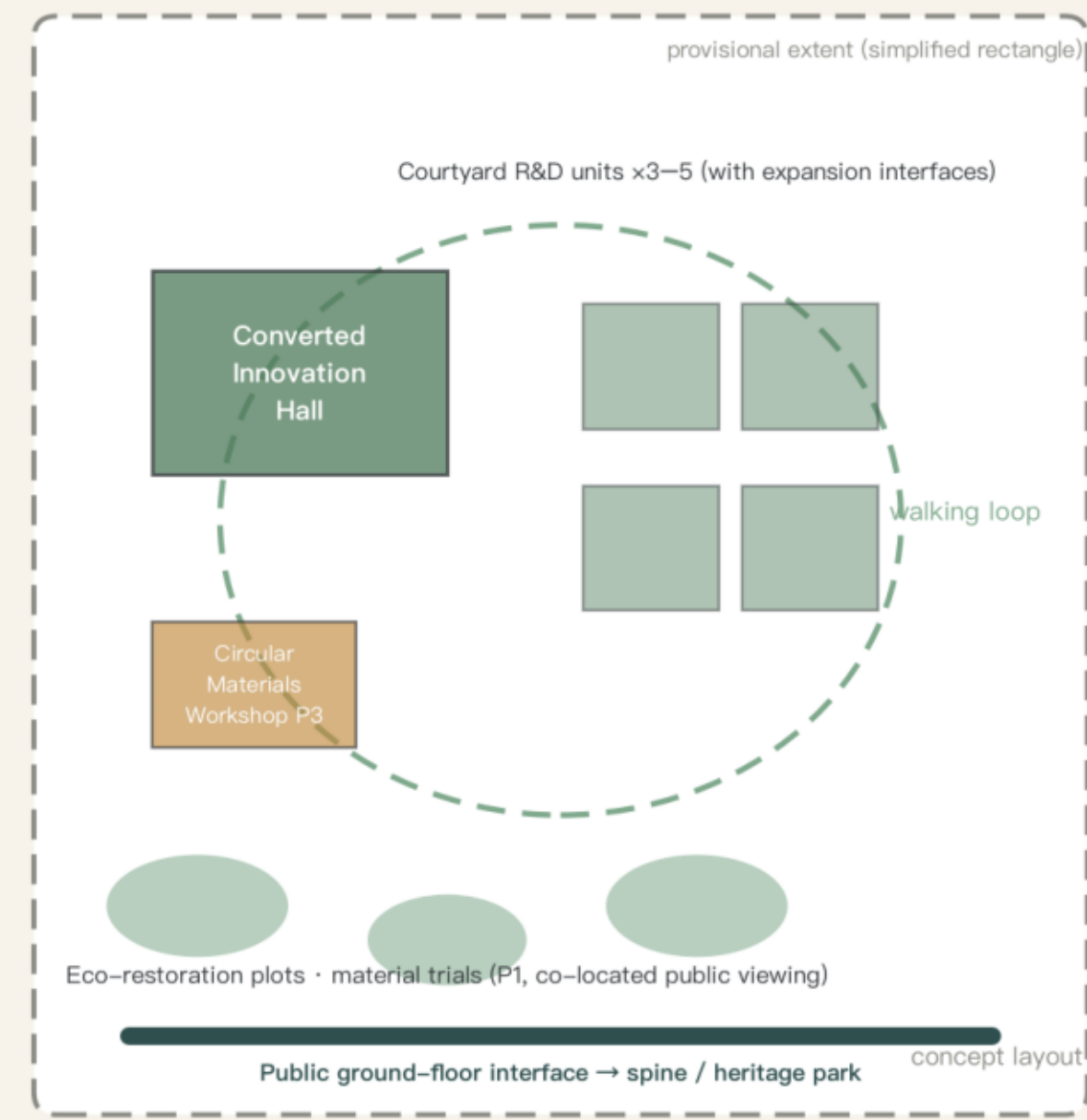
Statutory greening rate, FAR, and building height are regulatory control indicators; official values are not public and remain unknown — no numerical conclusions

03 Key–Area Detailed Design

Nurture — Transfer — Use | each column: concept layout + Jingzhang growth–section excerpt + use scenarios | layouts are conceptual, not master plans

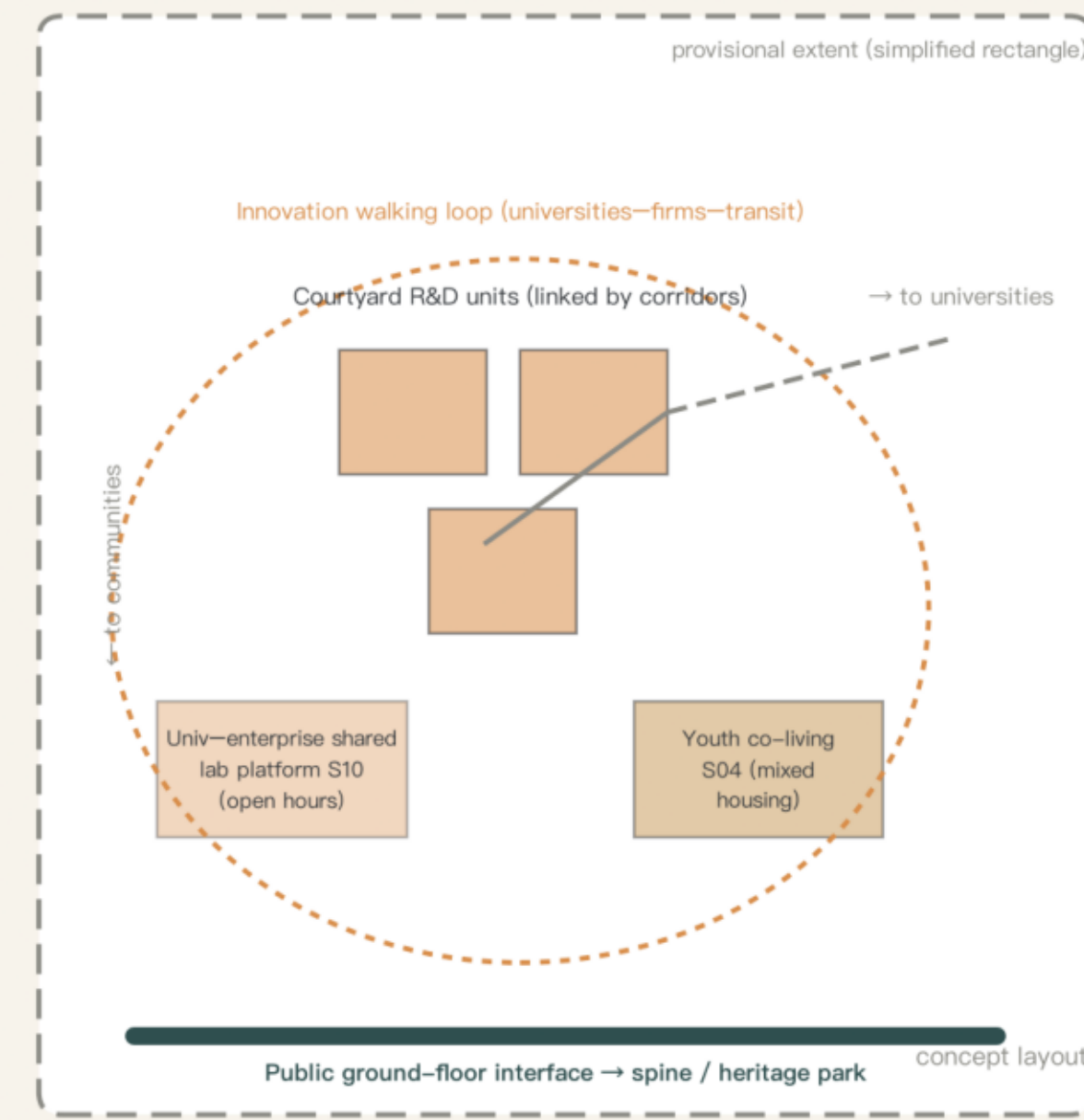
Zhongzhiyuan · Nurture

Garden–type AI innovation district | 192.1 ha (announced)



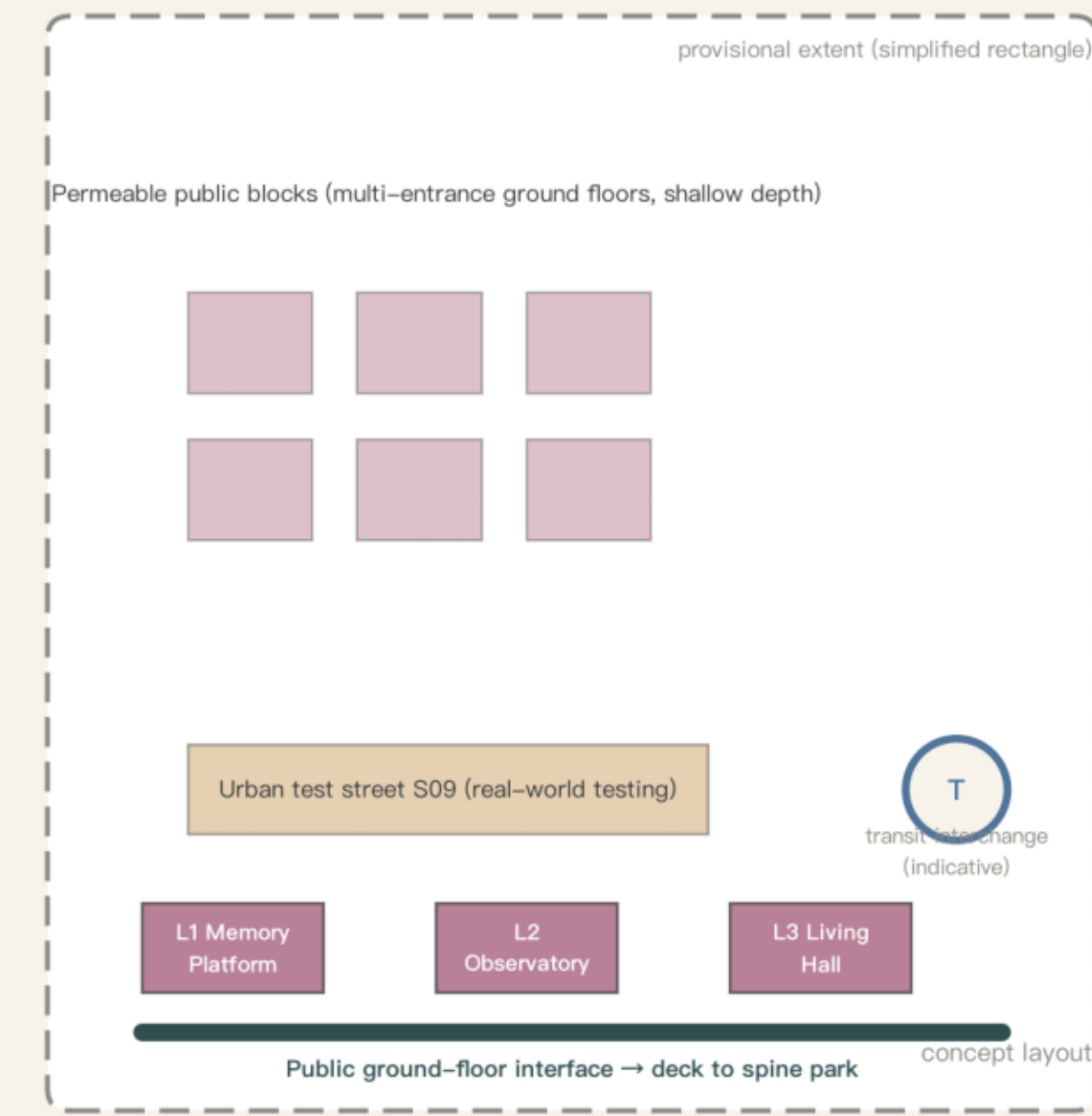
AI Origin · Transfer

Campus–adjacent AI innovation district | 104.3 ha (announced)



Dazhongsi · Use

Urban AI innovation district | 72.0 ha (announced)

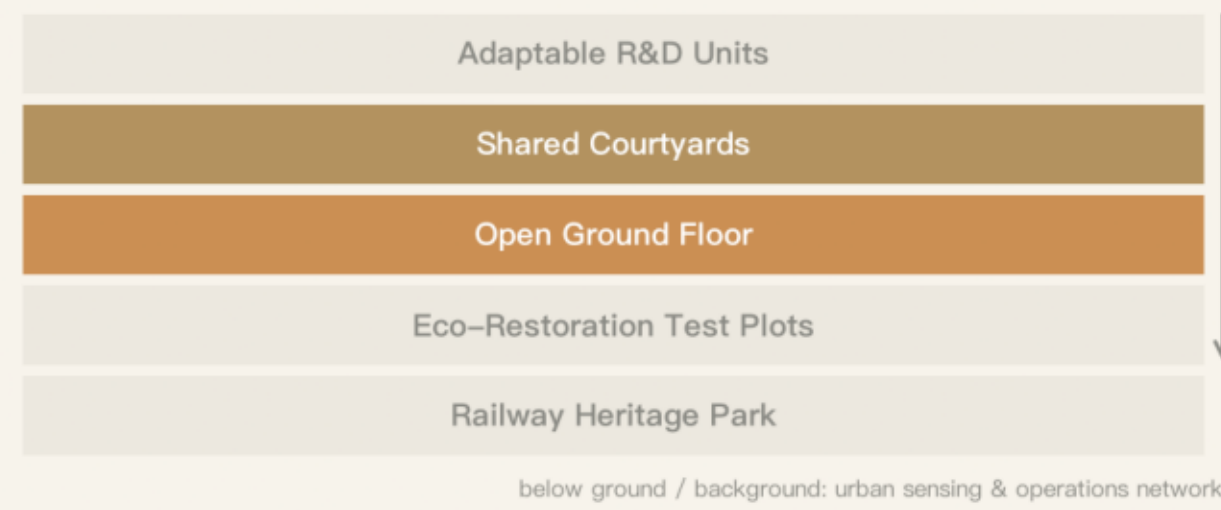


Growth section: all five layers (prototype area)



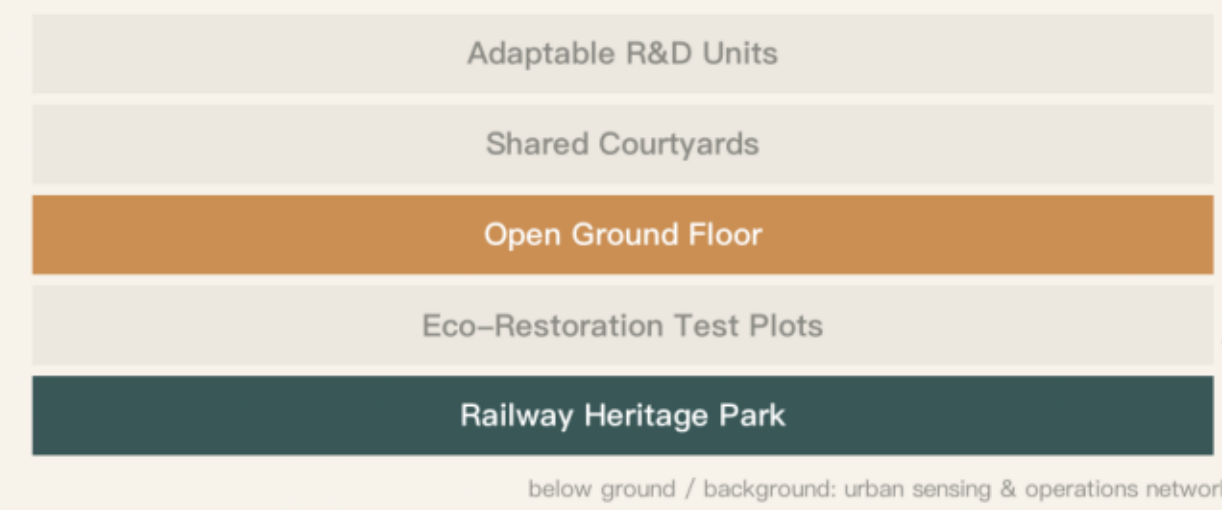
- Use scenarios:
- Weekday daytime: prototype teams in residence; plot monitoring co–located with public viewing
 - Night: shared canteen, sports and night–study facilities support resident teams
 - Weekend: public training in the materials workshop, guided tours and exhibits at restoration plots

Section excerpt: emphasis on courtyards–ground floor



- Use scenarios:
- Weekday daytime: shared–equipment slots, joint university–enterprise courses and experiments
 - Night: shared kitchens and night study in the youth co–living residences
 - Weekend: ground–floor public courses and exhibitions; community market on the innovation walking loop

Section excerpt: emphasis on ground floor–park



- Use scenarios:
- Rush hours: commuting transfers and community service stations tested by real passenger flows
 - Night: public events at the Memory Platform; heritage gallery open late
 - Weekend: shows at the future public living hall; public feedback days on the test street

Conceptual diagram; not an official boundary or planning basis. Boundaries are provisional geometry. Source: open–city–ai/haidian public task package

Fig. 3 Detailed design of the three key areas: positioning, conceptual master plans, spatial sequences, typical sections, and first–phase hooks; all boundaries are provisional geometry to be corrected once official red lines are released. This plate is a conceptual version.

Zhongzhiyuan AI Independent Innovation Acceleration Area · ~192.1 ha (announced)

Nurture — prototyping and validation ground: one Retrofitted Innovation Hall plus 3–5 Courtyard R&D clusters carry the full–stack R&D–testing–pilot chain; reserved potential land opens first as temporary green and test plots; a city–level stitching point strengthens station–park–axis links; Qinghe culture starts the heritage narrative in the north; test plots and trial fields become watchable productive landscape on the ground floor. First phase: P1 ecological baseline and restoration plots, P3 circular materials workshop and component library, S01 observation anchor station, one Courtyard R&D pilot, one station–axis stitching point (E04, E01).

Beijing AI Origin Community · ~104.3 ha (announced)

Transfer — open R&D and knowledge–sharing campus: shared lab platforms sit where campus boundaries meet the axis, with equipment open to community booking; Youth Co–Creation Residences hedge the innovation premium with low–cost retention; low–disturbance renewal through block–level stitching of walls, passages, and courtyard entrances; an innovation walking loop links universities–enterprises–station–axis; the former Qinghuayuan Station anchors the knowledge narrative. First phase: the first open unit of S10, the first converted S04 residence, 1–2 station–campus stitching points, the S07 archive launch (E01, E03).

Dazhongsi AI Industry Cluster · ~72.0 ha (announced)

Use — public–life testing and heritage–renewal living room: a Permeable Public Block carries display, trading, and experience of agent, terminal, and AI–content formats mixed with wet markets and convenience retail; continuous, shaded, barrier–free four–quadrant walking connections around the station; parking–occupied negative spaces gradually become public ground floors and slow–traffic space; the Urban Test Street receives prototypes from Zhongzhiyuan and AI Origin, and the observatory publishes corridor–wide baselines, results, and failure records. First phase: the first P2 station, the S05 transfer segment, the first S09 test–street segment, one four–quadrant stitching point, the L2 preparatory exhibition (E02, E06).

Scenario	Users	Spatial carrier	AI role & data boundaries
S01 Environmental Observation Station	Researchers, citizens, students	Public station + test plots (Zhongzhiyuan)	Anomaly detection, trend forecasting; non-identifying data, anonymised, published
S02 Urban Ecological Restoration Garden	Residents, students, O&M	Rain gardens, intervention-control plots (P1)	Data archiving, strategy simulation; no claims without control confirmation
S03 AI Community Service Station	Elderly, children, users with disabilities	Community halls, corner stations (Dazhongsi P2)	Guidance and form pre-filling; data minimisation, human fallback, opt-out
S04 Youth Co-Creation Residence	Young researchers, startups	Mixed housing + shared kitchens (AI Origin, Zhongzhiyuan)	Booking matching, energy optimisation; no behaviour scoring
S05 Accessible Mobility Assistant	People with impairments, all pedestrians	Transfer nodes and slow-traffic paths corridor-wide	Route guidance, obstacle detection; edge processing, no trajectory retention
S06 Low-Speed Delivery Corridor	Merchants, residents, operators	Logistics interfaces, designated test segments	Path scheduling, conflict avoidance; marked vehicles, safety-limited data
S07 Railway Memory Digital Archive	Citizens, visitors, researchers	Galleries, Memory Platform (L1), online archive (E05)	Oral-history transcription, photo retrieval; human verification, AI content labelled
S08 Circular Materials Workshop	Designers, craftspeople, students	Retrofitted factories, open workshops (P3/E04)	Component retrieval and matching; material passports traceable
S09 Urban Test Street	Enterprises, residents, authorities	Adaptable streets and squares (Dazhongsi)	Tested objects plus operational data; boundaries and exits publicised
S10 Shared Lab Platform	Universities, enterprises, learners	Innovation halls, R&D courtyards (E01)	Equipment scheduling; data segregation by safety tier
S11 International Innovation Forum	International institutions, teams, citizens	Future Public Living Hall (L3, Dazhongsi)	Bilingual interpretation, archiving; human editorial approval
S12 Four-Season Public Calendar	All citizens, visitors, O&M	Axis nodes, squares, the heritage park	Scheduling optimisation, anonymised statistics; no facial recognition

Every scenario also specifies its operating body, core metrics, and stop/exit conditions (see booklet pp.11–12 and Ch.6); threshold numbers stay blank until the baseline survey, confirmed and published by the review panel — no false precision.

Three falsifiable first-mover validation projects

P1 Main-axis ecological baseline and restoration plots (Zhongzhiyuan): a full growing-season baseline plus co-located intervention/control groups; falsification point — no meaningful difference after a full cycle means public acknowledgement and withdrawal from the toolkit.
P2 AI community services and accessibility public station (Dazhongsi): human and digital services in parallel; falsification point — vulnerable-group satisfaction or completion below the pure-human baseline means AI does not hold and the station reverts to human mode.
P3 Circular materials and demountable public components (Zhongzhiyuan): reused and new components side by side in the same facility; falsification point — higher maintenance cost or failure rates than control with no improvement path means scaling back to display use.
Unified protocol fields: purpose, space, baseline, control, responsible parties, continue conditions, stop conditions, restoration actions, falsification point.

Six personas & three signature carriers

Community residents / young researchers and students / startups and enterprise teams / universities and research institutions / children, the elderly, and users with disabilities / visitors and international partners (O&M staff as the implicit seventh group).
L1 Jingzhang Memory Platform — where we come from; L2 Urban Intelligence Observatory — how we understand the present; L3 Future Public Living Hall — where we go together. Signature quality comes from unique urban public functions, not overscaled form.

Implementation & operations

Six innovation-ecosystem project packages: E01 shared experimental facilities / E02 urban real-scenario testing / E03 youth residency and co-creation / E04 circular construction platform / E05 open heritage archive / E06 public-value procurement — mapped one-to-one to key areas, operating bodies, and public returns.
Four-stage implementation: 0 baseline and consensus (months 0–12) → 1 connect first (years 1–2) → 2 key areas (years 2–5) → 3 network expansion (after year 5); conceptual pacing, to be corrected once ownership, approvals, funding, and engineering conditions are clear.
Five-party structure: public platform governing body + professional operations team + university research consortium + community deliberation mechanism + enterprise project partners.
Six Admission Questions: real problem / space and public resources / visible public return / data and protection / continue-adjust-exit metrics / restoration after exit.
Long-term loop: annual public-event calendar, the Jingzhang Workshop community, two scenario launch windows with a permanent 'exit archive' exhibition, and bilingual international communication with open toolkits.

Annual public-event calendar (S12) — all events free or low-cost by default; the commercial share is capped and published in the annual operations report

Spring

Main-axis ecology observation season (public open days at test plots) · annual launch of new pilots

Summer

Youth co-creation season (residency works exhibition, summer public courses) · waterfront night events

Autumn

Heritage and memory season (archive collection, railway culture week) · annual forum (L3)

Winter

Evaluation and deliberation season (annual pilot review, community assembly, co-creation of the next calendar)

05 Phased Implementation & Evidence Dashboard

Four phases · three falsifiable pilot projects (protocol cards) · baseline–intervention–evaluation loop | verifiable metrics instead of visions

Four-Phase Implementation (conceptual pacing)

